KRISHNA KUMAR JHA

SOFTWARE ENGINEER |FULL-STACK / AI SYSTEMS

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TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Java, C++
Frontend: React, React.js, HTML, CSS, Streamlit
Backend: FastAPI, Node.js, Express, Spring Boot, REST APIs, JWT, Hibernate/JPA, Prisma
Data & Infrastructure: PostgreSQL, Redis, Docker, Kubernetes
Engineering: Distributed Systems, Async Processing, API Design, Authentication, Authorization, Rate Limiting, Validation, Error Handling, Testing, Git, GitHub Actions, Postman
AI/ML: Scikit-learn, Pandas, Feature Engineering, Model Deployment, Real-time Inference

PROJECTS

CodeFlow — Secure Code Execution Platform **GitHub**
Tech Stack: FastAPI, React, PostgreSQL, Redis, Docker

- Architected an asynchronous distributed execution pipeline where REST requests enqueue jobs in Redis and dedicated workers execute them in isolated Docker sandboxes, decoupling API latency from code execution.
- Built a security boundary for untrusted Python, Java, and C++ execution with network isolation, 128 MB memory caps, CPU quotas, 10-second timeouts, and controlled I/O limits; tested infinite loops, OOM conditions, and network-escape attempts.
- Implemented JWT authentication, rate-limited APIs, normalized PostgreSQL persistence, execution history/status APIs, and graceful output truncation; validated concurrent execution with 5 jobs without result cross-contamination.

Expense Tracking System — Production-Oriented Personal Finance Platform **GitHub**
Tech Stack: React.js, TypeScript, Node.js, Express, Prisma, PostgreSQL

- Built a feature-based three-tier application spanning React UI, Express middleware/controllers/services, Prisma data access, and PostgreSQL, covering expenses, categories, budgets, dashboards, reports, and exports.
- Implemented secure authentication with short-lived access JWTs, refresh-token rotation/storage, HTTP-only Secure SameSite cookies, bcrypt password hashing, ownership-scoped queries, validation, CORS, rate limiting, and centralized error handling.
- Designed transactional workflows, composite database indexes, pagination and sorting, and SQL-backed dashboard/report aggregation to support reliable and efficient production-style data access.

DevFlow — Developer Workflow & Project Management Platform **GitHub**
Tech Stack: React, Spring Boot, PostgreSQL, Docker

- Built a full-stack developer workflow platform combining project/task management with live GitHub repository activity, including repositories, commits, and pull requests.
- Designed a layered Controller → Service → Repository backend with REST APIs and JWT-based authentication/authorization, then containerized the application for consistent deployment.

EXPERIENCE

CodSoft — Machine Learning Intern May 2024 – July 2024 | Remote

- Developed and deployed a Heart Disease Prediction Dashboard using Scikit-learn, Pandas, and Streamlit for real-time risk assessment from user-provided health metrics.
- Performed data preprocessing, feature scaling, exploratory analysis, and supervised model training, then integrated the trained model into an interactive web application for real-time inference.

EDUCATION

Lovely Professional University, Jalandhar 2022–2026
B.Tech in Computer Science & Engineering | CGPA: 7.49/10
HM Education Center, West Bengal 2020–2022
Senior Secondary Certificate | Percentage: 80.3

CERTIFICATIONS & ACHIEVEMENTS

Certifications: DeepLearning.AI TensorFlow Developer Professional Certificate (Coursera, 2026); Docker and Kubernetes (Udemy, 2026)
Achievements: Solved 250+ LeetCode DSA problems; Qualifying team, Infineon Hackathon — adaptive decision-making solution